



Aror University of Art, Architecture, Design & Heritage Sukkur.

BS(Artificial Intelligence) Fall-2025 Application of Information & Communication Technologies

Course Title: Application of ICT
Course Code: CSC-102
Credit Hours: (2+1)
Course Instructor: Rashida Naz
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Description:

As we live in the Computer Age, most of our daily activities are influenced by the use of computers and information technologies. In fields such as science and technology, meaningful progress is almost impossible without ICT. Therefore, it has become essential for everyone to develop a solid understanding of computers and their applications. This course, designed for first-year students without a computer science background, introduces the fundamental concepts and applications of Information and Communication Technologies. It covers core components that provide the foundation for advanced courses in computer science and related disciplines.

Aims and Objectives:

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| • Understand the fundamental concepts of Information and Communication Technologies (ICT) and their applications in daily life, education, and industry. |
| • Analyze the components of a computer system (hardware, software, and operating systems) and assess their role in performance. |
| • Learn how communication networks and the internet function, including LAN, WAN, and wireless networks, and explore their applications in home and office environments. |
| • Identify common security threats in ICT systems and apply basic measures to protect data and communication. |
| • Explore emerging ICT trends such as Artificial Intelligence, Cloud Computing, and the Internet of Things (IoT). |



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Assessment:

S. No	Assessment Activities	Percentage	Total Activities
1.	Sessional: Quizzes/ Assignments (Quizzes & Assignments)	30%	8-10
2.	Mid Term Exam	30%	1
3.	Final Exam	40%	1

Course Learning Outcomes (CLOs):

No.	Course Learning Outcome	Domain	Level	Assessment Tool
C1	Understand the fundamental concepts of ICT and their applications in daily life, education, and industry.	C	2	Class Participation, Quizzes, Mid Exams, Assignments
C2	Analyze the components of a computer system (hardware, software, and operating systems) and assess their role in performance	C	3	Class Activity, Quiz, Assignments
C3	Explain how communication networks and the internet function (LAN, WAN, wireless) and apply them in home/office environments	C	3	Worksheets, Mid Exam, Final Exam
C4	Demonstrate emerging ICT technologies such as Artificial Intelligence, Generative AI, Prompt Engineering, Blockchain, and Cybersecurity, emphasizing their ethical and societal impact.	C	2	Presentation, Final Exam

Domains:

C=Cognitive, A=Affective, P=Psychomotor

Levels:

Cognitive = {1: Remembering, 2: Understanding, 3: Applying, 4: Analyzing, 5: Evaluating, 5: Creating}

Affective = {1: Receiving, 2: Responding, 3: Valuing, 4: Organizing, 5: Characterizing}

Psychomotor= {1: Imitation, 2: Manipulation, 3: Precision, 4: Articulation, 5: Naturalization}



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Course Outlines:

Weeks	LEC#	SUBTOPICS
Week No: 01	Lec: 01	1.1 Introduction to the World of Computers: ○ Introduction ○ Computer in your life ○ Data vs Information ○ Generation of Computers
	Lec: 02	1.2 Parts of a Computer System: ○ Hardware ○ Software ○ Users 1.3 Information Technology: ○ Data ○ Characteristics of IT ○ Functions of IT
Week No: 02	Lec: 03	2.1 System Unit: ○ Digital Data & Program Representation ○ Binary Numbering System ○ Bits & Bytes
	Lec :04	2.2 Inside the System Unit: ○ The Motherboard ○ Power Supply and Drive Bays ○ Processors (CPU, GPU) ○ Memory (RAM, ROM, Cache, Flash) ○ Cooling Components ○ Expansion Slots & Cards ○ Buses ○ Ports and Connectors
Week No: 03	Lec: 05	3.1 Motherboard: ○ Ports and Cables ○ Expansion Cards and Slots
	Lec: 06	3.1 Buses and Memory: ○ Buses ○ ROM ○ RAM



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Week No: 04	Lec: 07	4.1 Storage Systems: ◦ Characteristics ◦ Primary Storage ◦ Secondary Storage
	Lec: 08	4.2 Other Storage Types: ◦ Network and Cloud Storage ◦ Other Storage Systems
Week No: 05	Lec: 09	5.1 Input Devices: ◦ Keyboards ◦ Pointing Devices ◦ Scanners, Readers, Digital Cameras
	Lec: 10	5.2 Output Devices: ◦ Audio Devices ◦ Display Devices ◦ Printers
Week No: 06	Lec: 11	6.1 System vs Application Software: ◦ Differences ◦ Functions of OS ◦ Types of OS
	Lec: 12	6.2 Utility Programs: ◦ Functions ◦ Types of Utility Programs
Week No: 07	Lec: 13	7.1 Application Software: ◦ General Characteristics ◦ Types
	Lec: 14	7.2 Productivity Applications: ◦ Word Processing ◦ Spreadsheet ◦ Database ◦ Presentation Graphics ◦ Multimedia Software ◦ Others
Week No: 08	Lec: 15	REVISION
	Lec: 16	Midterm Examination



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Week No:09	Lec: 17	9.1 Databases ○ Introduction to Databases ○ Data, Information, and Tables ○ Components of DBMS ○ Database vs File System ○
	Lec: 18	9.2 Database Management Systems (DBMS) ○ Types of Databases ○ Functions of DBMS ○ DBMS Software Examples (MS Access, MySQL) ○ Role of Databases in ICT Applications ○
Week No:10	Lec: 19	10.1 Programming Languages ○ Intro to Programming Languages
	Lec: 20	10.2 Computer communication Concepts: ○ Meaning and Importance of Computer Communication ○ Components of Communication System ○ Sending Device, ○ Channel, ○ Receiving Device ○
Week No: 11	Lec: 21	11.1 Types of Networks: ○ Local Area Network (LAN) ○ Metropolitan Area Network (MAN) ○ Wide Area Network (WAN) ○ Wireless Networks (WLAN, WWAN) ○ Examples of Networks in Daily Life
	Lec: 22	11.2 Network Architecture and Topologies: ○ Client–Server and Peer-to-Peer Models ○ Network Topologies: Star, Ring, Bus, Mesh ○ Intranet and Extranet Concepts
Week No: 12	Lec: 23	12.1 Networking Devices: ○ Networking Devices: Modem, Router, Switch, Access Point, Network Card
	Lec: 24	12.2 Communication Media: ○ Wired Media: Twisted Pair, Coaxial, Fiber Optic ○ Wireless Media: Infrared, Radio, Microwave, Satellite Networking



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Week No: 13	Lec: 25	13.1 Network Security and Safe Communication: ○ Basic Concepts of Network Security ○ Common Threats: Virus, Malware, Phishing, Hacking ○ Safe Internet Practices and Communication Ethics
	Lec: 26	13.2 Introduction to Artificial Intelligence: ○ Applications of AI in daily ICT systems (voice assistants, smart devices, etc.) ○ Examples of AI in daily life ○ Types of AI (narrow / general)
Week No: 14	Lec: 27	14.1 Generative AI & Prompt Engineering: ○ What is Generative AI ○ Examples (ChatGPT, image AI) ○ What is a Prompt ○ Ethical and responsible use of Generative AI
	Lec: 28	14.2 Networking and Blockchain Integration: ○ What is Blockchain ○ How blockchain uses networks ○ Applications & benefits of blockchain
Week No: 15	Lec: 29	15.1 Introduction to Cybersecurity: ○ Definition and importance ○ Common threats ○ Basic protection methods
	Lec: 30	15.2 Ethics, Privacy & Emerging Trends ○ Data ethics & privacy in AI / blockchain ○ Emerging tech trends (IoT, edge, cloud)
Week No: 16	Lec: 31	REVISION
	Lec: 32	Final Examination



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Text Books:

1. Understanding Computers: Today and Tomorrow, Comprehensive, 15th Edition
by Deborah Morley & Charles S. Parker

Reference

Books:

1. Norton, P. (6th ed.). *Introduction to Computers*. McGraw-Hill Education.
2. Shelly, G. B., & Vermaat, M. E.. *Discovering Computers: Fundamentals – Your Interactive Guide to the Digital World*. Course Technology, Cengage Learning.

Web Tutorials:

- <https://www.techtarget.com/searchcio/definition/ICT-information-and-communications-technology-or-technologies>
- [What Is Artificial Intelligence \(AI\)? | Google Cloud](#)
- [Artificial intelligence - Wikipedia](#)
- [Artificial Intelligence \(AI\) Basics for Beginners | Learning Tree](#)
- [Prompt Engineer Course - Blockchain Council](#)
- [AI Prompt Engineering in Cybersecurity](#)



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